SECTION 1 – DESIGN A LOGICAL MODEL

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PROMPT 1:

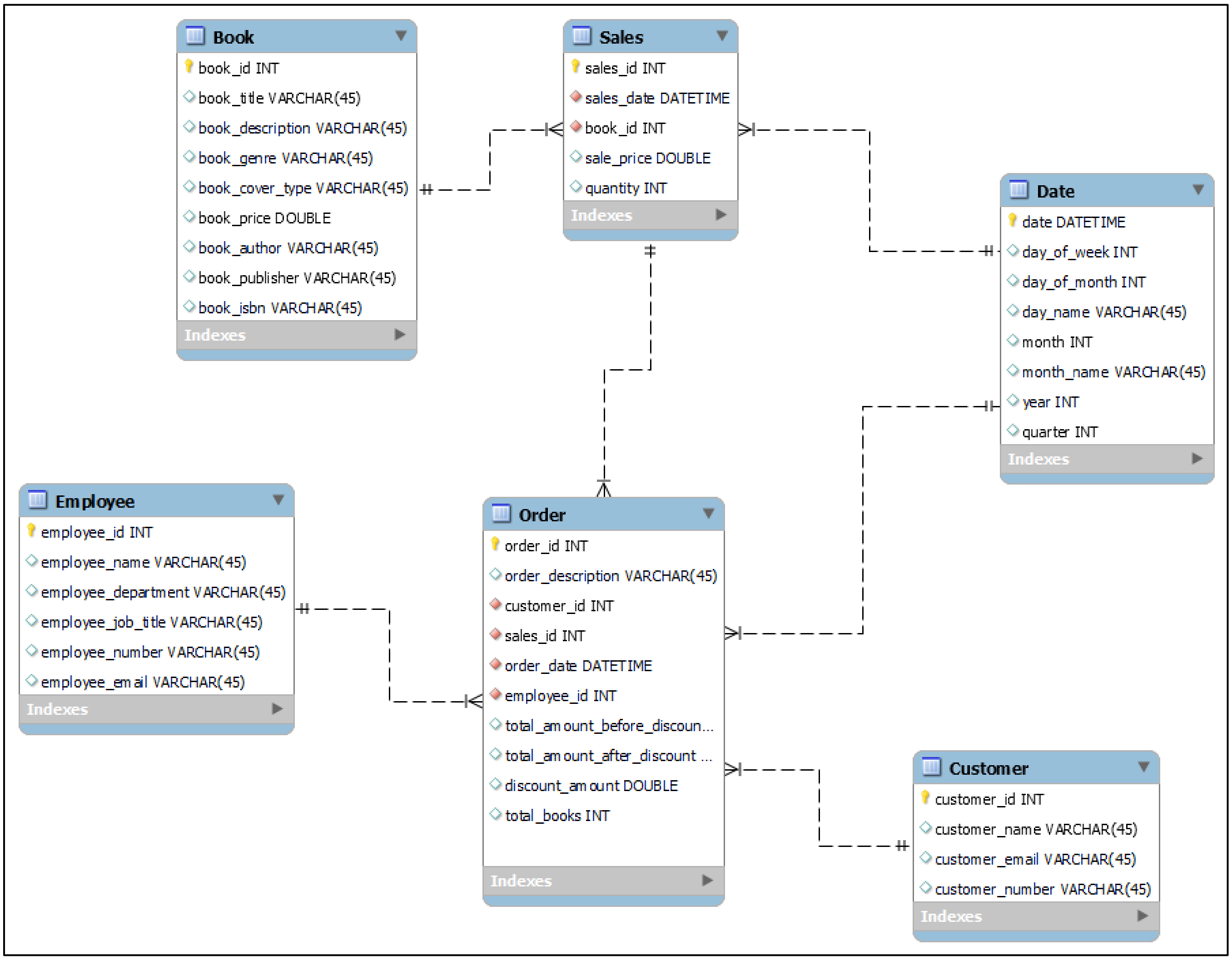

Figure 1 - ERD 1 with bookstore entities along with DATE table

PROMPT 2:

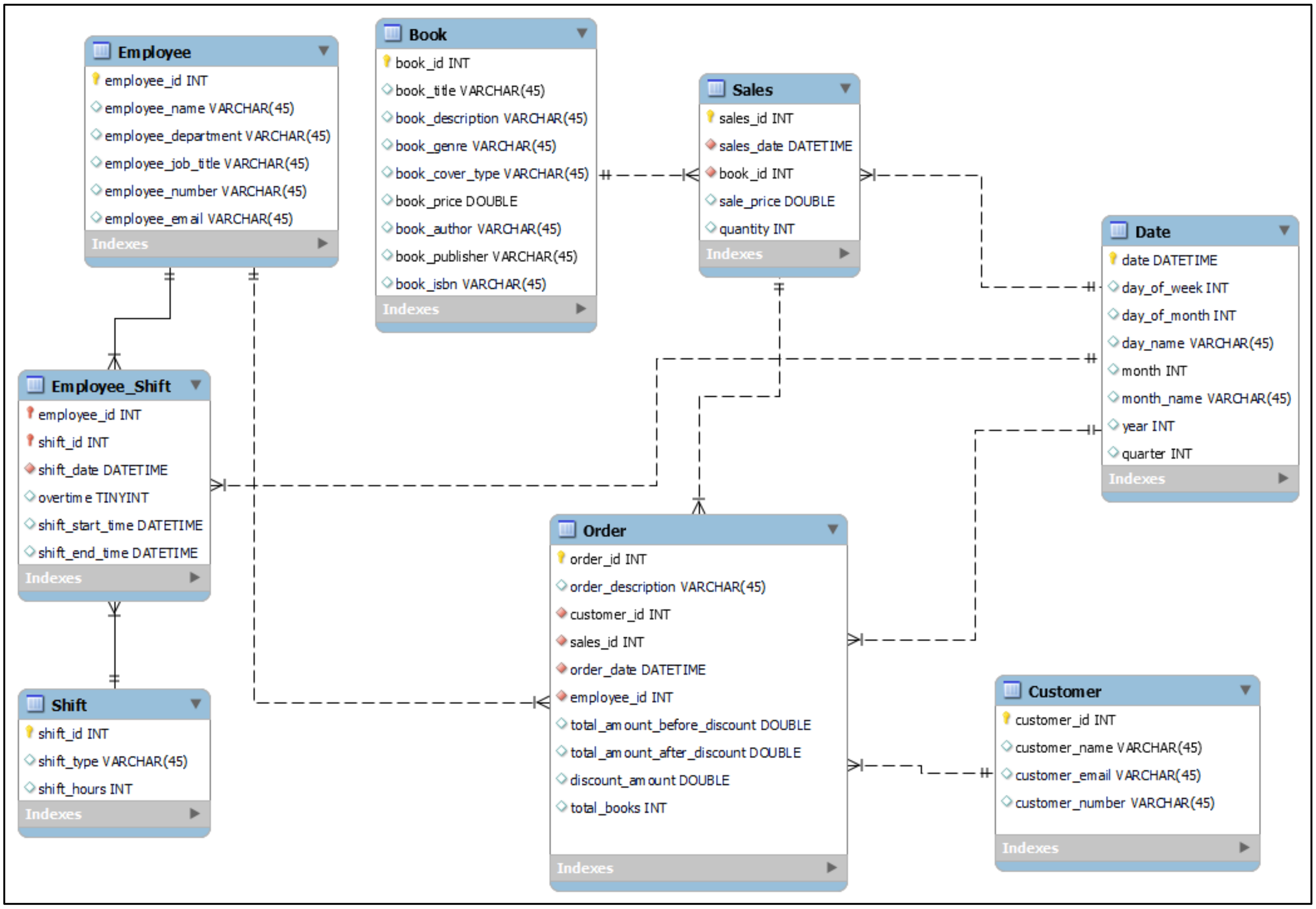

Figure 2 - Modified ERD with Employee Shift added - shift type in shift table represents (morning/evening)

PROMPT 3:

To store the customer address, we have two architecture proposals of slowly changing dimensions (SCD):

SCD Type 1:

In this type we will overwrite the address every time it changes so in this case the customer old address will be replaced with the new address any time there is a change in their address. To do this we simply add new columns that represent the address to the table as shown in figure 3 below.

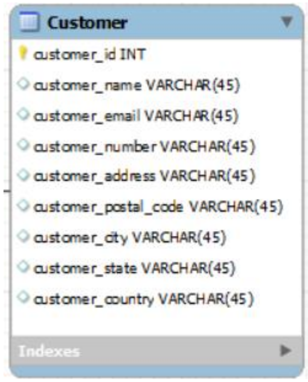

Figure 3 - customer address using SCD Type 1

SCD Type 2:

In this type we will retain the old address so we ensure the integrity of our data. In this example we won't lose the previous customer address because we will create a table that will hold all the historic data related to the address. The address table will have the customer as a foreign key. To know which address is the current one, we will add a Boolean column is_current_address and it will be true if the address is the most recent one and false otherwise as we can see in Figure 4 below.

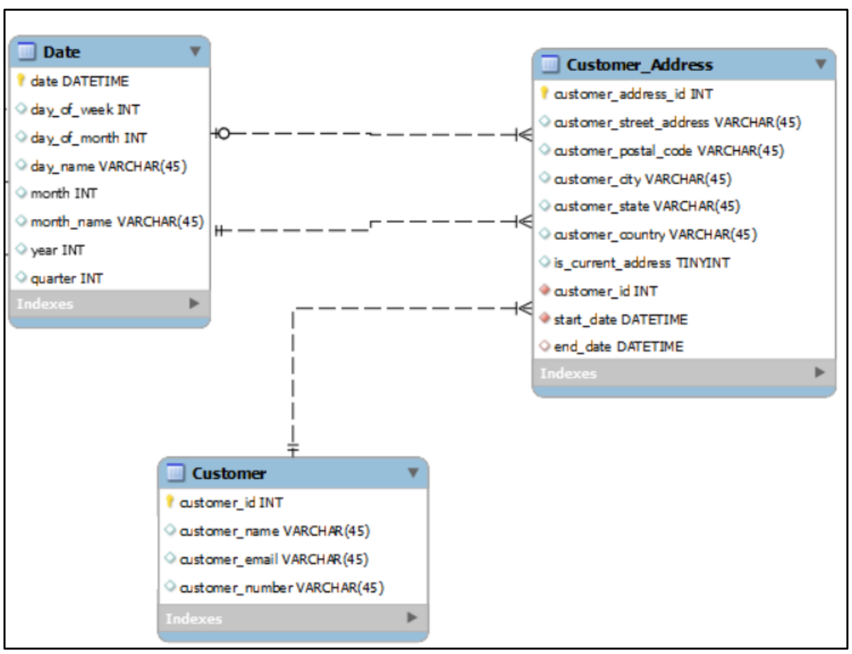

Figure 4 – SCD Type 2 creating a separate address table to retain its history